Referee Report

"Do Household Expectations Help Predict Inflation?"
by Brandao-Marques, Gelos, Hofman, Otten, Pasricha, and Strauss
(IMF Working Paper 2023/224; CEPR DP18774)

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1 Summary

Central banks track inflation expectations because what households believe about future prices shapes what they spend, save, and bargain for today. Most research looks at the median expectation. This paper looks at the whole distribution.

The authors use micro data from consumer surveys in the US, UK, Germany, and Canada. Section II documents how one-year-ahead expectation densities shifted right across all four economies after 2021. The econometric work narrows to the US, where the Michigan Survey of Consumers gives monthly data from 1978. The authors fit kernel densities, extract the median, variance, and skewness, and regress twelve-month-ahead CPI inflation on current inflation and those three moments. They also run a horse race against Consensus Forecasts and Cleveland Fed market-implied expectations.

Variance and skewness raise the in-sample R^2 over a model using only the median, though the gain concentrates almost entirely in high-inflation regimes identified by Bai–Perron (2003) breakpoint tests. The sign on variance switches: positive when inflation rises, negative when it falls. The authors read this through a staggered-updating lens in which better-informed households adjust first and others follow with a lag. Household moments retain predictive content in the 2017–2022 sub-period even after controlling for professional forecaster moments and market-based measures, though this rests on only sixty monthly observations. The paper draws on Mankiw, Reis, and Wolfers (2003) and Reis (2021). Its message is that central banks should watch the shape of the distribution, not just the median.

2 Major Concerns

1. The regime-dependence of the variance coefficient undercuts the forecasting claim.

The variance of household expectations predicts inflation. But the sign depends on whether inflation is rising or falling. A forecaster who wants to use disagreement as a signal must first know which regime she is in—the very thing the forecast is meant to tell her.

The authors know this. They do not solve it.

The breakpoints that define regimes come from a Bai–Perron test on the full sample. The regime labels embed information no forecaster had at the time. Table 6, the out-of-sample exercise without regime dummies, shows variance and skewness add nothing. The RMSEs improve only when the authors estimate on 1978–1990 and evaluate on 2020–2022 (Table 7). Both periods share high inflation. That is why it works. But it assumes the forecaster knows, after the fact, that the two episodes are alike. I also note an inconsistency: the text describes estimation on 1978M1–1990M7, while the Table 7 note states 1978M1–1980M12. If the latter is correct, the key cross-episode forecast exercise rests on three years of monthly data.

Two fixes would help. A Markov-switching model, or a threshold keyed to the trailing twelve-month CPI change, would let the sign of variance shift endogenously. The interaction terms in Table 3 move this direction but still condition on ex-post inflation thresholds. The authors should also report formal forecast-comparison tests for nested models with overlapping horizons, as discussed in point 4.

2. The baseline level regression may capture common persistence, not predictive information.

The authors estimate the main model in levels. A footnote acknowledges that unit roots in inflation and the moment series cannot be rejected over the full sample. They relegate first-difference results to an appendix. The significance weakens there. This goes to the heart of the paper's claim.

The strongest fit comes from the high-inflation regimes. Between them sits the Great Moderation, where the model explains almost nothing. When inflation, the median, variance, and skewness all rise together in high-inflation periods and stay flat in low-inflation periods, a levels regression finds significant coefficients. That reflects a shared low-frequency trend, not predictive content. Stock and Watson (2008) show that inflation-forecasting relationships are unstable across regimes.[1] Reis (2021) argues that shifts in the inflation anchor are slow, regime-like phenomena.[2]

The first-difference specification belongs in the main text. The authors should also report break-adjusted stationarity tests, a specification in deviations from trend or target, and rolling-window out-of-sample forecasts.

3. The two high-inflation windows are event-contaminated and not structurally comparable.

The paper's most striking out-of-sample result estimates the model on 1978–1990 and evaluates it on 2020–2022. This comparison is more fragile than the paper acknowledges.

The 1978–1990 window is not a clean high-inflation regime. It contains the second oil shock, the Volcker disinflation (the funds rate rose from roughly 11 to 19 percent), the 1980 and 1981–82 recessions, and the early Great Moderation. The negative variance coefficient in this window may simply reflect lagged household updating during the Volcker disinflation, not a general relation between disagreement and future inflation.

The 2015–2022 window is no cleaner. The Bai–Perron test labels 2015M11 as the start of a high-inflation regime, but 2015–2019 was a period of low inflation and falling oil prices. The actual surge arrived in 2021–2022, driven by COVID reopening, supply-chain disruptions, fiscal transfers, and the energy and commodity shocks that followed the Russian invasion of Ukraine. Household expectations during this period responded to highly visible everyday prices. The rightward shift of the household distribution may therefore reflect reactions to salient food and energy prices, not an independent predictive signal.[3]

These are not two draws from the same data-generating process. One is an unanchored-expectations, monetary-regime-change episode. The other is a pandemic supply-shock episode on top of decades of anchored expectations. That both produced high headline inflation does not justify transporting coefficients across them.

The authors should split each window into event-driven and non-event sub-periods, add controls for oil shocks, recessions, pandemic months, fiscal stimulus, and supply-chain disruptions, and run leave-one-episode-out tests. If the predictive content disappears outside the event windows, the paper should be reframed as evidence that household distributions react strongly during salient inflation episodes.

4. The paper has not run a forecasting horse race by the standards of the inflation-forecasting literature.

The current horse race is too narrow. Standard benchmarks include no-change and Atkeson–Ohanian average-inflation forecasts, AR models, Phillips-curve variants, energy-price-augmented

[1]Stock, J.H. and M.W. Watson, "Phillips Curve Inflation Forecasts," NBER Working Paper No. 14322, 2008.

[2]Reis, R., "Losing the Inflation Anchor," *Brookings Papers on Economic Activity*, Fall 2021.

[3]Federal Reserve History, "Oil Shock of 1978–79," "The Great Inflation," and "Recession of 1981–82"; Brookings Institution, "What Caused the U.S. Pandemic-Era Inflation?" 2023; BLS, "Exploring Price Increases in 2021 and Previous Periods of Inflation," *Beyond the Numbers*, 2022; D'Acunto, F., U. Malmendier, J. Ospina, and M. Weber, "Exposure to Daily Price Changes and Inflation Expectations," NBER Working Paper No. 26237, 2019.

models, and SPF forecasts.[4] The authors compare only against Consensus Forecasts and Cleveland Fed market-implied measures. Consensus Forecasts start in 1989, dropping the Volcker disinflation where the model fits best. The SPF runs from 1968.

The forecast evaluation method needs attention. Monthly data predicting twelve-month-ahead inflation produces heavily overlapping errors, and the median-only model nests inside the median-plus-moments model. The authors should use Clark–West or Clark–McCracken tests rather than relying on RMSE comparisons alone.[5] Without a proper contest against the established benchmarks, the paper has not shown that household moments help predict inflation.

5. The kernel bandwidth and the imputation rule receive no sensitivity analysis.

The authors take the bandwidth of 1.3 from Reis (2021) without discussion. Variance and skewness are tail statistics, and tail estimates from kernels are bandwidth-sensitive. The Michigan Survey codes "don't know how much up" as 96. The authors impute the mean of positive expectations. The share of these responses almost certainly moves with the inflation environment, since people give precise numbers less often when inflation is volatile. This injects measurement error correlated with the variation the regression exploits. Results under cross-validated bandwidths and alternative imputation rules (dropping code-96 responses, imputing the median) would tell me where the fragility lies.

3 Framing and Presentation

The paper states clearly that it discusses four countries and then zooms in on the US. The main regressions belong on US data. But the title and introduction still frame the paper as a general investigation. I reach Section III after the four-country plots in Section II expecting the cross-country thread to return. It does not. Running the same regression on the UK and Canadian quarterly data would show whether the result travels. If it does not, the title should say "Evidence from the United States."

The introduction also overpromises. Equation (1) is a forecasting regression, but the introduction invokes the New Keynesian Phillips Curve and household expectations shaping wage bargaining and consumption. A pure forecasting regression cannot speak to those channels. The introduction should match what the regression does.

4 Suggestions and Extensions

The conclusion notes that household expectations around turning points deserve further investigation. A probit or logit model using variance and skewness to forecast transitions into and out of high-inflation episodes would speak directly to the regime-switching problem in point 1. The Michigan Survey micro data carry demographics. If the predictive content comes disproportionately from younger cohorts or lower-income households, that tells us whether the signal reflects price pressures or the composition of who answers the survey.[6] A panel specification pooling the US, UK, and Canada would test whether the distributional channel operates similarly across central bank communication regimes. FPCA captures more of the distribution than three moments can; comparing its out-of-sample accuracy against the moments model would reveal how much information the summary throws away.

[4] Atkeson, A. and L.E. Ohanian, "Are Phillips Curves Useful for Forecasting Inflation?" *FRB Minneapolis Quarterly Review*, 25(1), 2001; Faust, J. and J.H. Wright, "Forecasting Inflation," in *Handbook of Economic Forecasting*, Vol. 2, Elsevier, 2013.

[5] Clark, T.E. and K.D. West, "Approximately Normal Tests for Equal Predictive Accuracy in Nested Models," *Journal of Econometrics*, 138(1), 2007; Clark, T.E. and M.W. McCracken, "Tests of Equal Forecast Accuracy and Encompassing for Nested Models," *Journal of Econometrics*, 105(1), 2001.

[6] Malmendier, U. and S. Nagel, "Learning from Inflation Experiences," *Quarterly Journal of Economics*, 131(1), 2016.

5 Recommendation

The paper has a good idea. The shape of the household expectations distribution moves with future inflation in ways that a staggered-information model predicts, the descriptive work is careful, and the theoretical motivation is clear.

The forecasting evidence, however, does not hold up yet. The two high-inflation windows are event-contaminated and not structurally comparable. The baseline level regression may reflect common persistence rather than predictive information. The variance coefficient changes sign across regimes, the regime labels use future information, and the horse race omits the standard benchmarks while lacking nested forecast evaluation tests. Data construction choices go unexamined.

Each of these has a fix. Split the high-inflation windows into event-driven and non-event sub-periods, with controls for oil shocks, recessions, and supply disruptions. First-difference and deviation-from-target specifications in the main text. A real-time regime classification scheme. A proper horse race against no-change, AR, Atkeson–Ohanian, Phillips curve, and energy-price benchmarks, with Clark–West or Clark–McCracken tests. Sensitivity checks on the bandwidth and the code-96 imputation. UK and Canadian regressions, or a title that says "Evidence from the United States."

I recommend revision and resubmission.